SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Analytical Problem Solving

Course Code:	ITA0532	Course Title:	Computer Vision
CO Assessed:	CO2 – Imitation (BL3)	Assessment:	Assessment Tool 1 – **Analytical Problem Solving**
Weightage:	10% of CIA	Total Marks:	100
CO1 Statement:	*Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

I ____________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __________________________

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Criteria	Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	Problem Understanding	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	Method Selection	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	Solution Process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	Accuracy of Calculation	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	Interpretation of Results	

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Question 1: Combined Enhancement & Segmentation

(a) Why is smoothing applied before segmentation?

A raw digital image almost always contains noise — small random variations in pixel intensity caused by the sensor, lighting, or transmission

Smoothing (for example, an averaging filter) blurs out this random noise by replacing each pixel with the average of its neighbourhood. This has three direct benefits for segmentation:

- It suppresses isolated noisy pixels so they don't get wrongly classified as object or background.
- It produces smoother intensity transitions, so the threshold boundary follows the real object edge instead of jumping around due to noise.
- It reduces over-segmentation — without smoothing, one real object can be broken into many small disconnected regions.

(b) Given image values — Averaging then Thresholding

Step 1: Take a 3×3 neighbourhood around the pixel to be smoothed:

$$\begin{matrix} 40 & 44 & 50 \\ 46 & 48 & 52 \\ 54 & 50 & 46 \end{matrix}$$

Step 2: Apply the averaging (mean) filter:

$$\text{Smoothed value} = \text{Sum of 9 pixels} / 9 = 432 / 9 = 48$$

This matches the given mean = 48.

Step 3: Apply thresholding with T = 50:

$$g(x,y) = 255 \text{ (object) if } f(x,y) >= T$$
$$g(x,y) = 0 \text{ (background) if } f(x,y) < T$$

Since the smoothed value 48 is less than T = 50:

$$48 < 50 \quad => \quad \text{Final Segmented Output} = 0 \text{ (Background)}$$

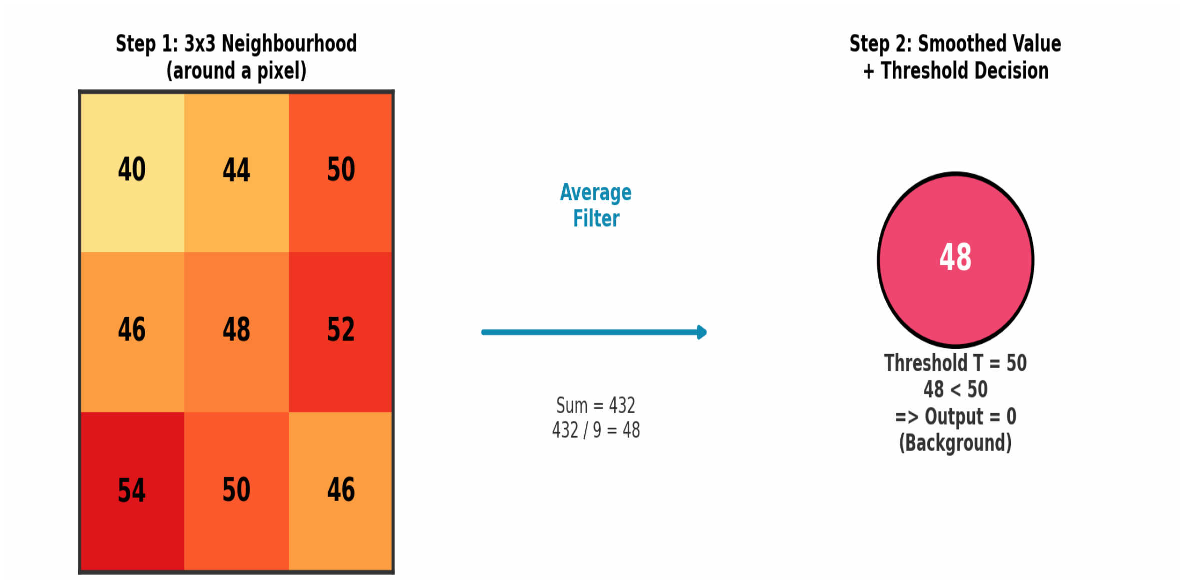


Fig 1: Step-by-step averaging and threshold decision

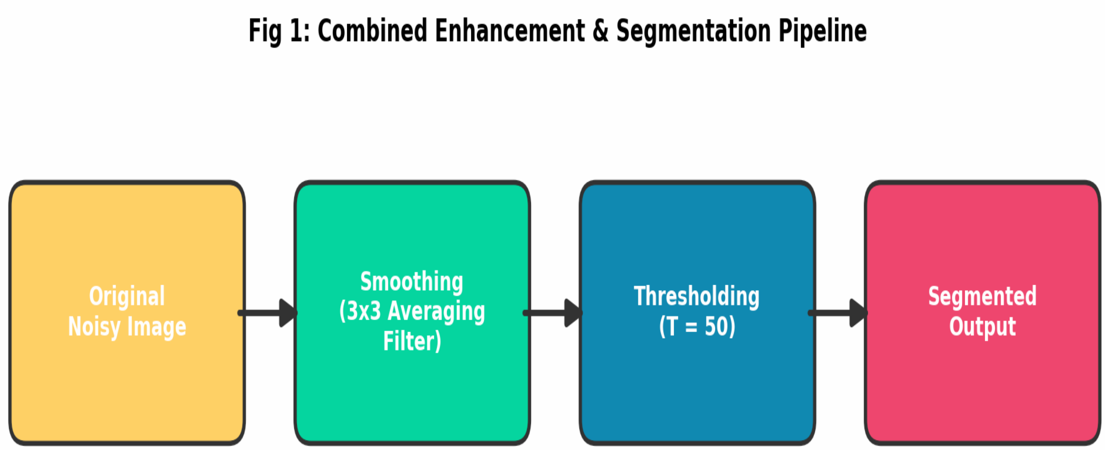


Fig 2: Overall combined enhancement and segmentation pipeline

Interpretation: The pixel, after smoothing, is dimmer than the threshold cut-off, so the segmentation stage labels it as background (0), not part of the object (255).

Question 2: Power-Law (Gamma) Transformation

(a) Role of gamma correction in image enhancement

Power-law (gamma) transformation reshapes the brightness of an image using a non-linear curve, instead of a straight line. It is used because display devices (monitors, cameras, scanners) do not respond to light linearly — they respond according to their own gamma characteristic.

$$s = c \cdot r^{gamma}$$

Here r is the input (normalised) intensity, s is the output intensity, c is a scaling constant (usually c = 1), and gamma controls the shape of the curve:

- gamma < 1 : the curve bends upward, mapping dark input values to much brighter output values -> the image becomes brighter, dark/shadow details become visible.
- gamma > 1 : the curve bends downward, compressing bright values -> the image becomes darker, useful for reducing washed-out highlights.
- gamma = 1: the transformation is the identity line -> no change.

This single family of curves lets one formula handle both brightening and darkening simply by changing one parameter, which is why it is preferred over a fixed linear formula.

(c) & (d) Step-by-step application and computation

assumed data: *Specific pixel values were blank in the question sheet. The commonly used test set r = {50, 100, 150, 200} with c = 1 and gamma = 0.5 (brightening case) is used below.*

Step 1: Normalise each pixel value to the range [0, 1] by dividing by 255:

Step 2: Apply s_norm = r_norm ^ gamma, then

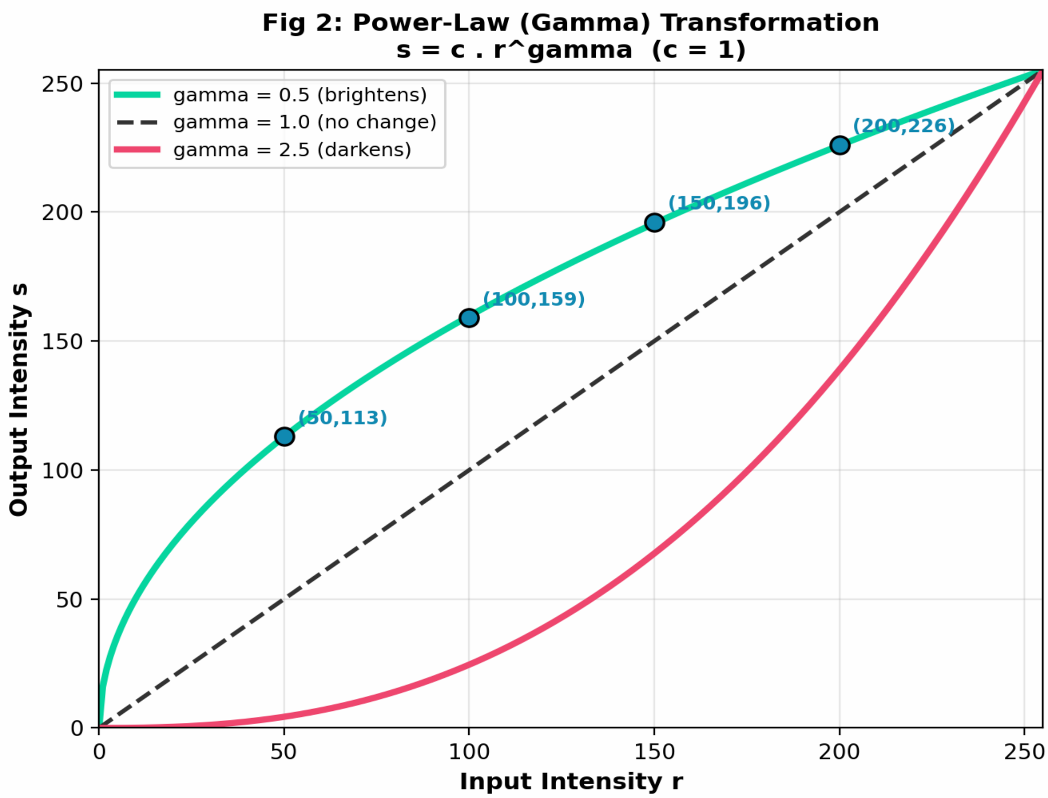
Step 3: scale back: s = s_norm x 255.

r (input)	r_norm = r/255	r_norm^0.5	s = s_norm x 255

50	0.196	0.443	113
100	0.392	0.626	159
150	0.588	0.767	196
200	0.784	0.886	226

Fig 3: Gamma curves (0.5, 1.0, 2.5) with the computed points marked

(e) Interpretation of results



Every output value is higher than its corresponding input value (e.g. 50 -> 113, 100 -> 159). This confirms that gamma = 0.5 brightens the image, with the effect being strongest for the darkest pixel (50 -> 113, more than double) and weakest for the brightest pixel (200 -> 226, a smaller relative jump). This non-uniform boost is exactly what gamma correction is meant to do: lift shadow detail without blowing out the highlights

Question 3: Contrast Stretching

(a) Need for contrast stretching

The image has all its intensity values squeezed into a narrow band, [50, 150], out of the full possible range [0, 255]. Such an image looks flat, hazy, and low-contrast on screen because it never uses the darkest blacks or the brightest whites

(b) Justifying the linear transformation to [0, 255]

$$s = (r - r_{min}) / (r_{max} - r_{min}) \times 255$$

Since the input range [50, 150] is a straight sub-interval of intensities, a simple linear (piecewise-linear) mapping is enough and is preferred because it is easy to compute, fully reversible in shape, and does not distort the relative order or spacing of intensities — it only rescales them. With $r_{min} = 50$ and $r_{max} = 150$, this formula guarantees that the darkest input (50) maps exactly to 0 and the brightest input (150) maps exactly to 255.

(c) & (d) Step-by-step application and computation

assumed data: *Specific pixel values were blank in the question sheet. The test set r = {50, 75, 100, 125, 150}, spanning the given range, is used below.*

$$s = (r - 50) / (150 - 50) \times 255 = (r - 50) \times 2.55$$

r (input)	(r - 50)	(r - 50) x 2.55	s (output, rounded)
50	0	0.00	0
75	25	63.75	64
100	50	127.50	128
125	75	191.25	191
150	100	255.00	255

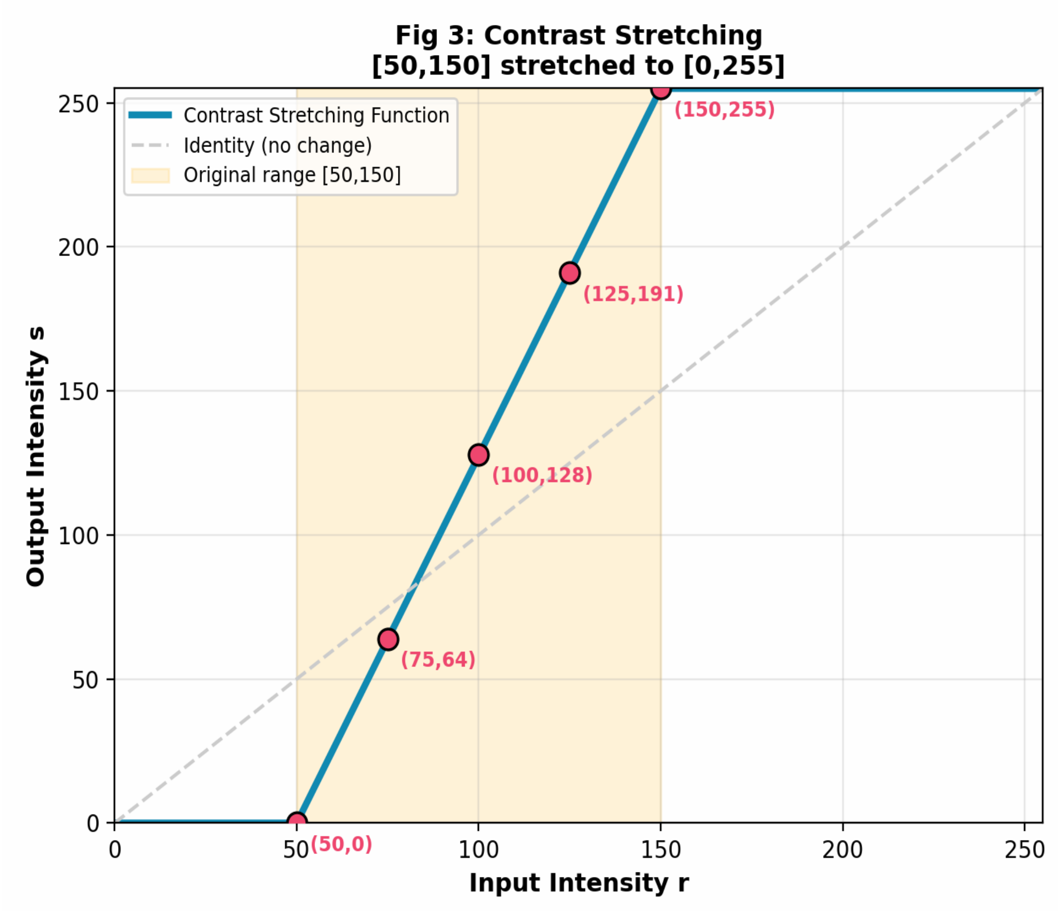


Fig 4: Contrast stretching function mapping [50,150] to [0,255]

(e) Interpretation of improvement in contrast

Before stretching, all five sample pixels were squeezed inside just 100 grey levels (50 to 150). After stretching, they are spread across the full 0-255 range with equal spacing preserved (0, 64, 128, 191, 255). The gap between adjacent samples grew from 25 to about 64 grey levels, meaning intensity differences that were barely visible before are now clearly distinguishable. The image therefore shows richer, punchier contrast without any change in the actual structure or content of the scene.

Summary Table — Key Results

Question	Method Used	Final Result
Q1(b)	3x3 Averaging + Threshold (T=50)	Smoothed value 48 -> Output = 0 (Background)
Q2(c-d)	Power-Law, gamma=0.5, c=1	50->113, 100->159, 150->196, 200->226 (brighter)
Q3(c-d)	Linear Contrast Stretch [50,150]->[0,255]	50->0, 75->64, 100->128, 125->191, 150->255